

Darren Lam

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PROFILE

- **Programming Languages:** Java, Python, JavaScript, TypeScript, HTML/CSS, C, C++, C#, ARMv8, Haskell, Prolog
- **Tools & Technologies:** Git, GitHub, React.js, React Native, Node.js, Express.js, Flask, JavaFX, SQL, SQLite, MySQL, PostgreSQL, MongoDB, Postman, Docker, Kubernetes, WebSocket.IO, JUnit testing, Visual Studio Code, IntelliJ, PyCharm, UNIX, Linux, Azure, Excel, Object-Oriented-Programming, CI/CD, DevOps, Google Cloud, TCP/UDP Protocols, NMAP, Wireshark

EDUCATION

Bachelor's of Science, Computer Science

September 2023 - Present

University of Calgary, Calgary, AB

- **Relevant coursework:** Data Structures and Algorithms, Software Engineering, Computer Networks, Object-Oriented Programming, Operating Systems, Database Management Systems, AI Research, Computer Graphics

TECHNICAL EXPERIENCE

Software Engineer Intern

May 2026 - Present

General Dynamics Mission Systems – Canada, Calgary, AB

- Developed software features using **C#** and **Python** based on project requirements in an **Agile** environment
- Automated and performed **manual testing** in Python, **peer validating** code changes before production releases
- Configured hardware and updated firmware to support **software integration** and **compatibility testing**

Software Engineer Intern

May 2025 - August 2025

ATCO, Calgary, AB

- Developed an early iteration of CollectPal in VS Code using Python, increasing collection efficiency by approximately **30%**
- Worked closely with field-workers to streamline CollectPal, resulting in the completion of **3,260** readings over a single month
- Collaborated with the Itron and Temetra development teams to deploy and test CollectPal in the field

Information Technology Intern

September 2023 - April 2024

CovarsaDx Corp., Downey, CA

- Migrated internal workflows to SharePoint, increasing collaboration efficiency by **25%** across **50+** users
- Resolved technical support requests with a **95%** first-contact resolution rate, reducing downtime by **75%** compared to last year's statistics
- Implemented cybersecurity measures, including extensive system monitoring and automated spam email disposal

PROJECTS

CollectPal

May 2025 - August 2025

A React.js/SQLite/Flask stack web application to assist with gas meter-reading collection

- Developed a **BFS-style GPS mapping algorithm** using **Google Maps API** and **haversine mathematics** to ensure **100%** accuracy in finding the most efficient collection route
- Incorporated a **SQLite and Flask backend system** to assist with responsive updates to the CSV-based GPS route, reducing load times by **25%** and increasing work organization by **50%**
- Implemented real-time **NMEA** location fetching using a **VK-162** GPS dongle device to provide the Maps API with accurate location data
- Incorporated a seamless web page that favours user ergonomics with **React UI**, and a streamlined front-end to back-end system

Lostly

September 2025 - December 2025

A blog-style web application constructed using React.js/MySQL/Flask/Socket.IO for a digital lost-and-found service

- Implemented a **Flask-based backend** providing **RESTful** endpoints and **Socket.IO** real-time communication
- Integrated **Flask-login** with secure **password hashing** for user authentication and stored user credentials on a **MySQL DB**
- Designed and seeded a normalized **MySQL schema** using relational model structure with **foreign keys** and **connection-retry logic** for robust DB access
- Built secure image **upload/server pipeline** (server-side validation, secure filenames), **CORS support**, and environment-driven configuration via **python-dotenv**

EXTRACURRICULARS

Suspension R&D Engineer

September 2024 - Present

Formula SAE UCalgary Racing, Calgary, AB

- Designed front control arm assemblies for the **UCR-02** car using SolidWorks CAD and Adams Car Dynamic Testing Software
- Stress-tested control arm assemblies inside SolidWorks via FEA, resulting in a final iteration with an overall FOS of **2.4** under **15,600N** of load
- Ran **FEA simulations** on all carbon fiber two-force link tubes using **Ansys Mechanical** for the **UCR-03** to determine appropriate sizing based on stress and FOS results